

4. Since atoms were found to contain fundamental particles, they must be divisible.
Hence, the correct option is (a).
5. $Z = \frac{29392}{1837} = 16$
Hence, the correct option is (b).
6. A dipositively charged helium ion is α particle. A helium atom has 2 electrons, 2 protons, and 2 neutrons. When 2 electrons are removed, it is left with 2 protons and 2 neutrons that means 4 fundamental particles.
Hence, the correct option is (b).
7. The presence of nucleus in an atom was first established by Rutherford.
Hence, the correct option is (a).
8. Platinum is also highly malleable and has high nuclear charge. So, it can replace gold foil in α -ray scattering experiment.
Hence, the correct option is (c).
9. An atom with $Z = 7$ and $Z = 15$ has electronic configurations 2, 5 and 2, 8, 5. These two can form trinegative ions.
Hence, the correct option is (d).
10. Noble gases or inert gases have 8 valence electrons.
Hence, the correct option is (c).
11. The electronic configuration of element X is 2, 8, 10, and 2. The atomic number of the element is 22.
Hence, the correct option is (b).
12. Electron releases energy when it moves from M shell to L shell, since M shell has greater energy than L shell.
Hence, the correct option is (a).
13. α -ray scattering experiment was conducted to prove the validity of Thomson's model. All the α -particles were expected to pass straight through the gold foil since the positively charged mass is uniformly distributed.
Hence, the correct option is (d).
14. Electronic arrangement of an element having atomic number 24 $\rightarrow$ 2, 8, 13, 1
K L M N
So, the antepenultimate shell has 8 electrons.
Hence, the correct option is (b).
15. (i) $\rightarrow$ (D); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A)
Hence, the correct option is (d).

CHAPTER 2

Classification of Matter

Assessment Test I

1. Water will be in vapour state because there is no external pressure.
Hence, the correct option is (c).
2. When we exhale air from our mouth on a winter morning, the warm water vapour present in the exhaled air condenses to form water droplets.
Hence, the correct option is (c).
3. Diamond is a very good conductor of heat because of the compact arrangement of small carbon atoms.
Hence, the correct option is (a).
4. Smoke is not a homogeneous mixture.
Hence, the correct option is (d).
5. Sodium is a soft metal and others are hard.
Hence, the correct option is (d).
6. The correct sequence is using separating funnel for separating benzene, then fractional distillation to separate alcohol and finally evaporation to obtain sugar.
Hence, the correct option is (b).
7. In position A, it will dry up faster because the rate of evaporation is directly proportional to both temperature and surface area.
Hence, the correct option is (d).
8. B can be liquefied by the application of pressure at that temperature, not A.
Hence, the correct option is (b).
9. Formation of ice from water and condensation of water vapour involve decrease in potential energy.
Hence, the correct option is (b).
10. Pressure cooker reduces the cooking time by increasing the boiling point of water due to increase in the external pressure.
Hence, the correct option is (b).
11. Metals have lustre because of transition of loosely bound electrons.
Hence, the correct option is (a).
12. Steel is more resistant to corrosion because alloying makes the constituent metal less reactive.
Hence, the correct option is (a).
13. Chromatogram (a) represents a mixture of R, B, and G dyes. Chromatogram (b) represents a mixture of R, B, and G dyes and some other impurity also. Chromatogram (c) represents a mixture of red and green dyes. Chromatogram (d) represents a mixture of red and blue dyes.
 $\therefore$ The order is BDAC.
Hence, the correct option is (b).
14. At first, sand has to be filtered. CCl_4 has to be separated by using separating funnel. Then, fractional crystallization is done to separate KNO_3 . To separate NaCl , evaporation is required.
 $\therefore$ The order is DCBA.
Hence, the correct option is (d).
15. (i) Copper sulphate + water $\rightarrow$ Solid-liquid homogeneous mixture
(ii) Benzene + water $\rightarrow$ Liquid-liquid heterogeneous mixture
(iii) Fine powdered sulphur + water $\rightarrow$ Solid-liquid heterogeneous mixture
(iv) Vinegar + water $\rightarrow$ Liquid-liquid homogeneous mixture
(v) Steel $\rightarrow$ Solid-solid homogeneous mixture
Hence, the correct option is (c).

Assessment Test II

1. Ice kept in a vacuum chamber sublimates and gets converted to vapour state.
Hence, the correct option is (c).

2. Cooking gas (LPG) is liquefied by the application of pressure.
Hence, the correct option is (b).
3. It is graphite, and so the constituent atoms are carbon.
Hence, the correct option is (c).
4. Brass, being an alloy of copper, is a homogeneous mixture.
Hence, the correct option is (a).
5. Among the given substances only gas carbon is a good conductor of electricity.
Hence, the correct option is (c).
6. At first, charcoal should be separated by filtration. Then coconut oil can be separated by using separating funnel and glucose can be obtained by evaporation.
Hence, the correct option is (a).
7. We blow air to cool a hot liquid in order to provide fresh air above the surface of the liquid.
Hence, the correct option is (c).
8. Critical temperature of C is the highest because it can only be liquefied at that temperature.
Hence, the correct option is (c).
9. A cup of hot tea at 60°C gets cold.
Hence, the correct option is (b).
10. During skating, pressure is applied on ice and it melts, and during filling up of cooking gas cylinder also, pressure is applied to liquefy the cooking gas.
Hence, the correct option is (d).
11. Metals are good conductors of electricity because of the presence of free electrons. Metal atoms have generally 1 or 2 or 3 electrons in their valence shell.
Hence, the correct option is (c).
12. Duralumin is as strong as steel because it is an alloy of aluminium, and alloying improves

the mechanical properties of metal. Alloying reduces the malleability of the metal.

Hence, the correct option is (c).

13. Chromatogram (a) contains A, B, and two other components. Chromatogram (b) contains A, C, and another component. Chromatogram (c) contains A, B, and C. Chromatogram (d) contains A and C.

∴ The order is BCAD.

Hence, the correct option is (c).

14. At first, gravity separation can be carried out to separate sawdust. Then filtration can be done to remove coarsely ground chalk, followed by fractional distillation for removing vinegar. Then evaporation can be done to obtain the common salt.

∴ The order is DBAC.

Hence, the correct option is (a).

15. (i) Solid–solid homogeneous mixture → Bronze
(ii) Solid–liquid homogeneous mixture → Saline water
(iii) Liquid–liquid heterogeneous mixture → Coconut oil and water
(iv) Liquid–liquid homogeneous mixture → Petrol and diesel
(v) Solid–liquid heterogeneous mixture → Finely powdered charcoal in water.

Hence, the correct option is (a).

Assessment Test III

1. The thermal expansion is maximum in gases.
Hence, the correct option is (c).
2. In both evaporation and boiling, liquid changes to its gaseous state.
Hence, the correct option is (c).
3. Good conductance of electricity is the characteristic of a metal. Non-metals are generally poor conductors of electricity.
Hence, the correct option is (a).

4. Phosphorus – 4

Iodine – 2

Sulphur – 8

Ozone – 3

∴ The increasing order of atomicity is BDAC.

Hence, the correct option is (a).

5. (i) German silver → Copper

(ii) Tungsten steel → Iron

(iii) Monel metal → Nickel

(iv) Duralumin → Aluminium

∴ The required order is CDAB.

Hence, the correct option is (b).

6. Fractional crystallization is used since both sodium chloride and potassium nitrates are soluble in water and have different solubilities.

Hence, the correct option is (d).

7. Potassium has the least density.

Hence, the correct option is (a).

8. Since alcohol and water are miscible, they cannot be separated by using a separating funnel. They have to be separated by fractional distillation.

Hence, the correct option is (c).

9. If intermolecular force of attraction of a gas is more, then its critical temperature is higher. Critical temperature is the temperature above which it is not possible to liquefy a gas by applying pressure. That means, critical temperature is proportional to the intermolecular force of attraction. Between CO_2 and O_2 , CO_2 has greater intermolecular force of attraction and hence, greater critical temperature.

Hence, the correct option is (a).

10.

(i) $\text{NH}_4\text{Cl} + \text{KNO}_3$	One of the components can sublime on heating
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(ii) Sulphur + carbon disulphide	Non-sublimable solid solute in liquid solvent.
(iii) Soda water	Decrease in solubility of a gas in liquid with the decrease in pressure.
(iv) Air	Difference in the boiling points of components in their liquid state.

Hence, the correct option is (a).

11. The higher is the vapour pressure, the lower is the intermolecular forces of attraction. The lower is the intermolecular force of attraction, the lower is the critical temperature. Therefore, A has the least vapour pressure.

Hence, the correct option is (a).

12. The boiling point of liquid B is greater than that of A since a liquid boils at a temperature when its vapour pressure becomes equal to the atmospheric pressure.

Hence, the correct option is (b).

13. Milk is a heterogeneous mixture as it is a colloid.

Hence, the correct option is (a).

14. Sea water, because it contains maximum percentage of dissolved salts.

Hence, the correct option is (c).

15. Skating of ice is possible because increase in pressure decreases the melting point of ice. This is because ice is a solid, which contracts on melting.

Hence, the correct option is (b).

Assessment Test IV

1. If the same amount of heat is supplied, then the iron block shows the least expansion since it is a solid substance.

Hence, the correct option is (a).

2. Boiling is a fast process. $\therefore$ Option (d) follows.
Hence, the correct option is (d).
3. Mercury can conduct electricity in pure state since it is a metal.
Hence, the correct option is (b).
4. The atomicity of helium is 1. Hydrogen, chlorine, and nitrogen have atomicity 2.
Hence, the correct option is (c).
5. In bell metal, the major component is copper.
Hence, the correct option is (c).
6. (i) Solvent extraction $\rightarrow$ Sand is insoluble whereas sugar soluble in water.
(ii) Chromatography $\rightarrow$ A technique used for the separation of colours of pigments.
(iii) Magnetic separation $\rightarrow$ Iron is magnetic whereas iodine is nonmagnetic.
(iv) Sublimation $\rightarrow$ Naphthalene is sublimable whereas common salt is not.
Hence, the correct option is (c).
7. (i) O_2 gas $\rightarrow$ Gas has the least density.
(ii) Ice $\rightarrow$ Ice has less density.
(iii) Water $\rightarrow$ Water has greater density.
(iv) A solid block of aluminium $\rightarrow$ Solids have maximum density.
Hence, the correct option is (d).
8. Although NH_4Cl dissolves in water, it sublimes on heating. So, it cannot be separated by distillation.
Hence, the correct option is (b).
9. NH_3 is an easily liquefiable gas because of its high intermolecular forces of attraction.
Hence, the correct option is (c).

10.

(i) Distillation	Non-sublimable solid solute in liquid solvent and can be obtained after separation.
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(ii) Filtration	Insoluble solid component present in a liquid.
(iii) Preferential liquefaction	Difference in the ease of liquefaction of the component gases.
(iv) Separating funnel	Immiscible liquids present in distinct layers.

Hence, the correct option is (b).

11. Easily liquefiable gas has greater intermolecular force of attraction. Intermolecular force of attraction of A is more than that of B.
Hence, the correct option is (c).
12. The higher is the vapour pressure, the lower is the boiling point. The liquid with lower vapour pressure boils at a higher temperature, whereas the liquids having higher vapour pressure have higher intermolecular force of attraction.
Hence, the correct option is (c).
13. Air is a homogeneous mixture since it is a mixture of various gases.
Hence, the correct option is (b).
14. Presence of impurities increases the boiling point of a liquid.
Hence, the correct option is (a).
15. Increases in pressure increases the melting point for solids which expand on melting. For solids which contract on melting, increase in pressure decreases the melting point for solids.
Hence, the correct option is (a).

CHAPTER 3

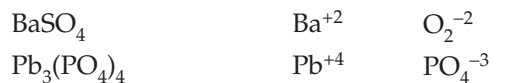
Language of Chemistry and Transformation of Substances

Assessment Test I

- Formula of chloride of trivalent metal M is MCl_3 .
 $\therefore$ Balanced chemical equation is $\text{MCl}_3 + 3\text{NH}_4\text{OH} \rightarrow \text{M}(\text{OH})_3 + 3\text{NH}_4\text{Cl}$.
 Hence, the correct option is (d).
- Valency of non-metal X in XH_3 with respect to H is 3. Given that valencies of X in the two oxides are 2 and 4 with respect to oxygen, and formulae of oxides are XO and XO_2 .
 Hence, the correct option is (a).
- IC and per acids of chlorine are HClO_3 and HClO_4 , respectively.
 $\therefore$ The formulae of salts with trivalent metal are $\text{M}(\text{ClO}_3)_3$ and $\text{M}(\text{ClO}_4)_3$. The number of oxygen atoms in these salts, respectively, are 9 and 12.
 Hence, the correct option is (a).
- Carbon forms two oxides CO and CO_2 , and the valencies of carbon are 2 and 4, respectively.
 $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$ carbonic acid
 $\text{NaOH} + \text{H}_2\text{CO}_3 \rightarrow \text{NaHCO}_3$ (or) Na_2CO_3
 Hence, the correct option is (b).
- Both Pb and Sn form two oxides PbO , PbO_2 and SnO , SnO_2 , respectively. In PbO and SnO , the valencies of Pb and Sn are two, and in PbO_2 and SnO_2 the valencies of Pb and Sn are four.
 Hence, the correct option is (c).
- A can form A^+ ion and B can form B^{+2} ion.
 C can form C^{+3} ion and D can form D^{+4} or D^{+6} ion.
 $\therefore$ P is A_2O , Q is BO , R is C_2O_3 , and S is DO_2 or DO_3 .
 Hence, the correct option is (a).
- $\text{NaOH} + \text{HClO}_4 \rightarrow \text{NaClO}_4 + \text{H}_2\text{O}$
 (salt 1)
 $\text{Ca}(\text{OH})_2 + 2\text{HClO}_3 \rightarrow \text{Ca}(\text{ClO}_3)_2 + 2\text{H}_2\text{O}$
 (salt 2)
 $\text{Pb}(\text{OH})_4 + 4\text{HClO}_2 \rightarrow \text{Pb}(\text{ClO}_2)_4 + 4\text{H}_2\text{O}$
 (salt 3)
 $\text{Fe}(\text{OH})_3 + 3\text{HOCl} \rightarrow \text{Fe}(\text{OCl})_3 + 3\text{H}_2\text{O}$
 (salt 4)
 Hence, the correct option is (a).
- $\text{CaCO}_3 \xrightarrow{\text{Heat}} \text{CaO} + \text{CO}_2 - \text{Energy}$
 (X)
 $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{Energy}$
 (Y)
 Formation of X is endothermic and decomposition.
 Formation of Y is exothermic and compound–compound combination.
 Hence, the correct option is (a).
- Valency of metal M is 1 and metal ion is M^{+1} .
 Anion formed by non-metal A is A^{-2} .
 $\therefore$ Formula of compound is $\text{M}^{+1}, \text{A}^{-2} \Rightarrow \text{M}_2\text{A}$.
 Hence, the correct option is (b).
- Metal with electronic configuration 2, 8, 14, 2 is Fe. Fe can form two hydroxides $\text{Fe}(\text{OH})_2$ and $\text{Fe}(\text{OH})_3$. The formulae of salts of these hydroxides with sulphuric acid are FeSO_4 and $\text{Fe}_2(\text{SO}_4)_3$.
 $\text{Fe}(\text{OH})_2 + \text{H}_2\text{SO}_4 \rightarrow \text{FeSO}_4 + \text{H}_2\text{O}$
 $\text{Fe}(\text{OH})_3 + 3\text{H}_2\text{SO}_4 \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 3\text{H}_2\text{O}$
 Hence, the correct option is (b).

11.

Compound	+ve radical	–ve radical
$\text{Cu}_3(\text{PO}_4)_2$	Cu^{+2}	PO_4^{-3}
$\text{Sn}_2(\text{SO}_3)_4 = \text{Sn}(\text{SO}_3)_2$	Sn^{+4}	SO_4^{-2}



Hence, the correct option is (d).

12. Given that 100 g of metal oxide contains 60 g of metal.

$\therefore$ 40 g of metal oxide contains ? g of metal.

$$= \frac{40 \times 60}{100} = 24 \text{ g of metal}$$

$\therefore$ Atomic weight of metal is 24 and the metal is Mg.

$\therefore$ Molecular weight of $\text{Mg}(\text{ClO}_4)_2 = 24 + 71 + 128 = 223$

Hence, the correct option is (b).

13. The ratio by weight of oxygen that combine with fixed weight of nitrogen in NO , NO_2 , and N_2O_3 is $1:2:3/2 = 2:4:3$.

Hence, the correct option is (a).

14. Formula of ferric hydrogen phosphate is $\text{Fe}_2(\text{HPO}_4)_3$.

$$\therefore \text{Ratio} = \frac{2}{2+3+3+12} = \frac{2}{20} = \frac{1}{10} = 1:10$$

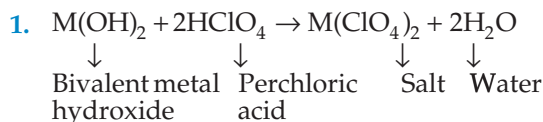
The valency of $\text{HPO}_4 - 2$ is 2.

Hence, the correct option is (c).

15. More reactive metal displaces less reactive metal from its aqueous solution. The order of reactivity of Zn, Fe, and Ni is $\text{Zn} > \text{Fe} > \text{Ni}$.

Hence, the correct option is (c).

Assessment Test II



Hence, the correct option is (a).

2. Valency of Y in H_2Y with respect to hydrogen is 2. The oxides of Y in which the valencies of Y differ by two units are YO_2 and YO_3 .

Hence, the correct option is (b).

3. The monovalent +ve radical with four hydrogen is AH_4^{+1} and bivalent -ve radical with four oxygen atoms BO_4^{-2} .

$\therefore$ The formula of the salt is $\text{AH}_4^{+1}, \text{BO}_4^{-2} = (\text{AH}_4)_2\text{BO}_4$.

Hence, the correct option is (d).

4. The formula of acid is H_3XO_4 .
 $\text{H}_3\text{XO}_4 + \text{M}(\text{OH})_2 \rightarrow \text{M}_3(\text{XO}_4)_2 + \text{H}_2\text{O}$

Hence, the correct option is (c).

5. Copper can form two oxides Cu_2O and CuO in which the valency of Cu is 1 and 2, respectively. Similarly Hg can form two oxides Hg_2O and HgO in which the valency of Hg is 1 and +2, respectively.

Hence, the correct option is (a).

6. P is A_2O_3 , Q is AO , R is A_2O_5 , and S is A_2O .

Hence, the correct option is (a).

7. Carbonate $\rightarrow \text{CO}_3^{-2}$, sulphate SO_4^{-2} , chlorite $\rightarrow \text{ClO}_2^{-1}$, and Hydroxide $\rightarrow \text{OH}^{-}$

Hence, the correct option is (c).

8. $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$

Hence, the correct option is (a).

9. Valency of X in XH_2 is 2 and X can form is X^{+2} . Y is P and it can form P^{-3} ion. The formula is $\text{X}^{+2}, \text{P}^{-3} \rightarrow \text{X}_3\text{P}_2$.

Hence, the correct option is (b).

10. Cations formed by metal M are M^{+1} and M^{+2} .

Formulae of nitrates are MNO_3 , $\text{M}(\text{NO}_3)_2$.

Formulae of nitrides are M_3N , M_3N_2 .

Hence, the correct option is (a).

11. Mercuric oxide (HgO) $\rightarrow 2$

Sodium peroxide (Na_2O_2) $\rightarrow 4$

Ferrous sulphate (FeSO_4) $\rightarrow 6$

Sodium phosphate (Na_3PO_4) $\rightarrow 8$

Hence, the correct option is (d).

12. The molecular weight of CaCO_3 is 100.

Hence, the correct option is (c).

13. In N_2O , N_2O_5 , and NO , the ratio by weight of oxygen that combine with fixed weight of nitrogen is $1:5:2$.

Hence, the correct option is (d).

14. In a compound, the total +ve charge is equal to that of -ve charge, but the total number of atoms in +ve ion may or may not equal to that of -ve ion.

Hence, the correct option is (c).

15. Metals which are more reactive than hydrogen on treatment with dilute acids gives hydrogen gas. The order of reactivity of Zn, Al, and H is $\text{Al} > \text{Zn} > \text{H}$.

Hence, the correct option is (c).

Assessment Test III

1. Burning of a candle involves:

- (i) melting of wax which is a physical change and
- (ii) combustion of hydrocarbons which is a chemical change.

Hence, the correct option is (a).

2. Compound is $\text{Ca}^{+2}, \text{P}^{-3} \Rightarrow \text{Ca}_3\text{P}_2$.

In a compound, the total positive charge and the total negative charge are equal.

Hence, the correct option is (d).

3. (a) In Zn_3P_2 , valency of positive radical is 2.
 (b) In Fe_2O_3 , valency of positive radical is 3.
 (c) In PbCl_4 , valency of positive radical is 4.
 (d) In NaNO_3 , valency of positive radical is 1.

Hence, the correct option is (b).

4. Neutralisation reaction $\Rightarrow$ (C), Thermal decomposition $\Rightarrow$ (B), Precipitation reaction $\Rightarrow$ (A), and Photolysis $\Rightarrow$ (D)

Hence, the correct option is (b).

5. **Compound** **Formula**

Sodium chlorate $\Rightarrow \text{NaClO}_3$

Sodium chlorite $\Rightarrow \text{NaClO}_2$

Sodium nitrate $\Rightarrow \text{NaNO}_3$

Sodium nitrite $\Rightarrow \text{NaNO}_2$.

Hence, the correct option is (b).

6. $\text{S} + 2\text{H}_2\text{SO}_4 \rightarrow 3\text{SO}_2 + 2\text{H}_2\text{O}$ is a balanced equation.

Hence, the correct option is (d).

7. Metal sulphate = $\text{M}^{+3}, \text{SO}_4^{-2} = \text{M}_2(\text{SO}_4)_3$

$$\therefore \text{Molecular mass} = 2(27) + 3(32) + 12(16) = 342$$

Hence, the correct option is (b).

8. The given reaction is a synthesis as well as a combination reaction.

Hence, the correct option is (c).

9. $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$

32 g of O_2 gives 2(30) g of NO.

? g $\leftarrow$ 45 g of NO

$$= \frac{32 \times 45}{60} = 24 \text{ g}$$

Hence, the correct option is (b).

10. In Na_2O , valency of Na = 1.

In ZnO , valency of Zn = 2.

In Fe_2O_3 , valency of Fe = 3.

In PbO_2 , valency of Pb = 4.

Hence, the correct option is (d).

11. In SO_2 , 32 g of S reacts with 32 g of oxygen.

In SO_3 , 32 g of S reacts with 48 g of oxygen.

$\therefore$ Ratio by weight of oxygen = 32:48 = 2:3.

Hence, the correct option is (b).

12. Atomic mass of Al is 27; atomic mass of S is 32; atomic mass of Ca is 40; atomic mass of O is 16.

Hence, the correct option is (c).

13. Symbol of lead is Pb.

Hence, the correct option is (c).

14. Na_2SO_4 is a salt of sodium hydroxide and sulphuric acid.

Hence, the correct option is (c).

15. Decomposition of C_2H_2 requires sound energy.

Hence, the correct option is (b).

Assessment Test IV

1. $\text{NH}_4\text{Cl} \xrightarrow{\text{Physical Change}} \text{NH}_4\text{Cl} \xrightarrow{\text{Chemical Change}} \text{NH}_3 + \text{HCl} \downarrow$

Hence, the correct option is (d).

2. The compound as a whole is neutral. Hence, the charges of +ve and -ve ions are neutralized by criss-cross method.

Hence, the correct option is (a).

3. (A) Charge on fluoride ion is -1.

(B) Charge on oxide is -2.

(C) Charge on nitride ion is -3.

(D) Charge on carbide is -4.

Hence, the correct option is (a).

4. The order of reactivity is $A > B$; $C > A$; $B > D$

$\Rightarrow C > A > B > D$

Hence, the correct option is (a).

5.

Compound	Total +ve charge
(i) Zn_3N_2	$3(+2) = +6$
(ii) PbCl_4	$1(+4) = +4$
(iii) Na_3PO_4	$3(+1) = +3$
(iv) $\text{Ca}(\text{NO}_3)_2$	$1(+2) = +2$

Hence, the correct option is (d).

6. Balanced chemical equation is $6\text{HNO}_3 + \text{S} \rightarrow \text{H}_2\text{SO}_4 + 6\text{NO}_2 + 2\text{H}_2\text{O}$.

$\therefore a = 6$ and $b = 2$

Hence, the correct option is (a).

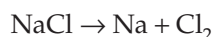
7. Metal phosphate = $\text{M}^{+2}, \text{PO}_4^{-3} = \text{M}_3(\text{PO}_4)_2$

Molecular mass = $a(3) + 2(31) + 8(16) = 262$

$\Rightarrow a = 24$

Hence, the correct option is (a).

8. The decomposition of a compound into its respective elements is known as analysis.



Hence, the correct option is (b).

9. $2\text{S} + 3\text{O}_2 \rightarrow 2\text{SO}_3$

(2×32) g of sulphur gives (2×80) g of SO_3 .

40 g of sulphur gives ? g of SO_3 .

$$= \frac{40 \times 160}{64} = 100 \text{ g}$$

Hence, the correct option is (a).

10. Fe shows variable valency.

Hence, the correct option is (a).

11. $\text{N}_2\text{O} \Rightarrow 28$ g of N and 16 g of O $\Rightarrow 140$ g of N and 90 g of O

$\text{NO} \Rightarrow 14$ g of N and 16 g of O $\Rightarrow 70$ g of N and 80 g of O

$\text{N}_2\text{O}_5 \Rightarrow 28$ g of N and 80 g of O

$\therefore$ Ratio of weight of nitrogen = 140:70:28

= 10:5:2

Hence, the correct option is (b).

12. Atomic mass of Mg is 24; atomic mass of S is 32; atomic mass of Na is 23; atomic mass of Cr is 52.

Hence, the correct option is (a).

13. Greek name of antimony is stibnum.

Hence, the correct option is (d).

14. Hydrochloric acid is a binary acid.

Hence, the correct option is (b).

15. Photosynthesis takes place in the presence of light; hence, it is a photochemical reaction.

Hence, the correct option is (a).

CHAPTER 4

Air and Oxygen

Assessment Test I

- In stratosphere, the temperature increases due to the increase in altitude.
Hence, the correct option is (b).
- 76 cm of Hg $\rightarrow$ 101292 Pascal
 $\therefore$ 19 cm Hg $\rightarrow$?
 $= \frac{1,01,292 \times 19}{76}$ Pascal = 25,323 Pascal
Hence, the correct option is (c).
- Since water occupies one-fifth of the volume of the inverted jar, it can be concluded that one-fifth of the volume of air is oxygen.
Hence, the correct option is (d).
- Photosynthesis does not consume O_2 .
Hence, the correct option is (a).
- Burning of petrol is an example of rapid combustion.
Hence, the correct option is (b).
- The innermost zone of a candle flame is the coldest zone.
Hence, the correct option is (a).
- Fe forms a mixed oxide, i.e., Fe_3O_4 ($FeO + Fe_2O_3$) on reaction with O_2 .
Hence, the correct option is (d).
- When a mixture of NH_3 and O_2 in 4:5 ratio is passed over platinum gauze at $800^\circ C$, NO and H_2O are formed.
Hence, the correct option is (a).
- $H_2S + O_2 \rightarrow H_2O + SO_2$
Hence, the correct option is (a).
- Surgical instruments are manufactured from stainless steel which is an alloy of iron.
Hence, the correct option is (d).
- Since combustion is an exothermic reaction, oxygen is used for the combustion of fuel

because it is a good supporter of combustion.
Hence, the correct option is (b).

- Since air is a mixture of different gases and one of the components of air is oxygen, air supports combustion.

Hence, the correct option is (a).

- The order of pollutants is DCAB.

(D) Usage of electrostatic precipitators in industries.

(C) Developing a suitable substituent for the refrigerant used in A/C, refrigerator, etc.

(A) Usage of scrubber in industries

(B) Controlling deforestation

Hence, the correct option is (d).

- The required order is DCBA.

(D) NO

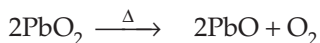
(C) CO_2

(B) NO_2

(A) P_2O_5

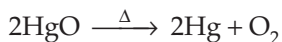
Hence, the correct option is (b).

- (i) $\rightarrow$ (D); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B)

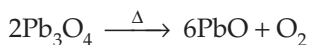


Chocolate
brown

Yellow
litharge



Mirror-like
surface near
the cooler parts of
the test tube



Reddish brown residue
which turns yellow on
cooling